RIVET GUN

1. Description

A rivet gun (rivet tool) is used to join two or more materials permanently using rivets. It is commonly used in sheet metal work, automobile repair, and fabrication labs.

Types of rivet guns:

- Manual (hand riveter)
- Pneumatic rivet gun
- Hydraulic rivet gun

In labs, a **manual hand rivet gun** is most commonly used. It works by pulling the mandrel, expanding the rivet body to lock materials together.

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2. How to Operate

Procedure:

1. Select correct rivet size (diameter and length).
2. Drill hole in the workpieces.
3. Align materials properly.
4. Insert rivet into the hole.
5. Insert rivet mandrel into rivet gun nozzle.
6. Hold gun straight (90° to surface).
7. Squeeze handles until mandrel breaks.
8. Remove the tool and check joint tightness.

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3. Do’s and Don’ts

Do’s:

- Use correct rivet size.
- Ensure proper alignment before riveting.
- Keep tool clean and dry.
- Use proper nosepiece for rivet size.
- Wear safety equipment.

Don’ts:

- Do not use damaged rivets.
- Do not tilt the rivet gun while operating.
- Do not apply excessive force unnecessarily.
- Do not use wrong size nozzle.
- Do not leave tool in dirty or wet conditions.

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4. Safety Measures

- Wear safety goggles to protect eyes from flying mandrel.
- Keep fingers away from moving parts.
- Ensure firm grip while squeezing handles.
- Do not point rivet gun towards others.
- Work in a stable and clean environment.

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5. Maintenance

Proper maintenance ensures long life and safe operation of the rivet gun.

Routine Maintenance:

- Clean the tool after every use to remove metal chips and dust.
- Wipe with dry cloth to prevent rust formation.
- Store in a dry place.

Lubrication:

- Apply light machine oil to moving joints and pivot points regularly.
- Avoid over-lubrication (prevents dirt accumulation).

Inspection:

- Check nosepiece for wear or damage.
- Inspect handles and joints for looseness.
- Ensure mandrel collection chamber (if present) is emptied.

Replacement:

- Replace worn-out nosepieces.
- Replace damaged springs or handles immediately.

Storage:

- Keep tool in toolbox or rack.
- Avoid dropping or mishandling.

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6. Real Image